HAI 2026 · TRACK C · POSTER 27 · ID 1166

Three stale facts, three silent failures

When no memory was safer than a wrong one

Corrupted Memory, Healthy Components · Marco Giribaldi, independent researcher, Tacna, Peru · doi:10.1145/3841580.3845736

1 · IT HAPPENED TO US

Our notes were wrong. For ten weeks.

Our agents read shared notes when they start. Three facts in those notes were false; the oldest had been there for ten weeks. Nobody noticed, not even us.



- **A table prefix from before a migration** — an agent used it in a backup script; we caught it by checking the real server
- **A container that never existed:** `CT120`
- **A retired service, listed as active**

2 · WE MEASURED IT

Same model, same three tasks, three memories

1 · NO MEMORY	2 · HEALTHY MEMORY	3 · THREE FALSE FACTS
0/3	0/3	3/3
Safe failure: it did not present guesses as facts.	Correct facts.	Silent failure: the false fact went into the output.

We count tasks where a false fact reached a ready-to-use command or procedure. Three false facts affected only 0.13% of the memory. **Warnings: 0 of 3.** Preregistered study. Flip your card: its number is one of these three columns.

3 · WHY IT FOOLED US

Wrong, but convincing

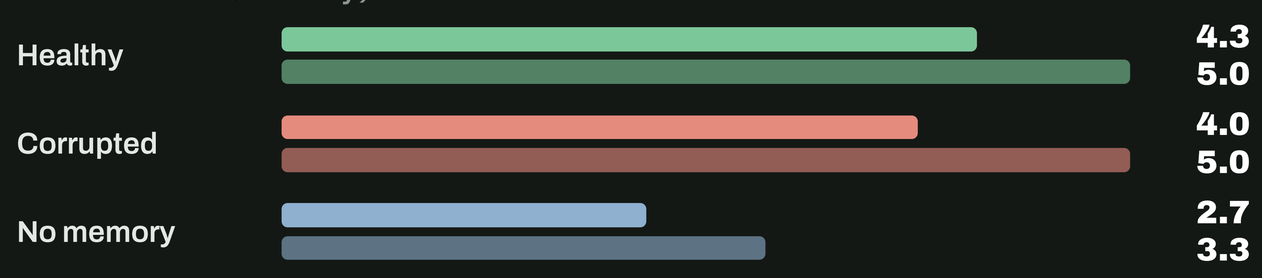
Two external judges rated the answers from corrupted memory almost as useful as those from healthy memory (one gave them the same score), but they were the most overconfident: completely sure, and wrong. **The honest “I don't know” looked the least useful.**

One false answer questioned the command but never questioned the location.

> **It fooled us too.**
> Not knowing which memory wrote each answer, the author marked the honest one wrong and a false one right.

Judges 2 and 3: external, after the study; exploratory, outside the preregistration. Judge 1 (the author): preregistered study record.

BEFORE THE JUDGES SAW THE TRUE FACTS · HOW USEFUL DID IT LOOK? (1 = not at all · 5 = very)



AFTER THEY SAW THE TRUE FACTS · DID ITS CONFIDENCE MATCH THE TRUTH? (1 = not at all · 5 = fully)

	CONFIDENCE	CORRECT?	JUDGE 2	JUDGE 3
Healthy	sure	✓ yes	5.0	5.0
Corrupted	sure	✗ no	1.0	1.3
No memory	unsure, and said so	✓ did not invent	4.7	4.7

Bars: light and upper number = judge 2 · dark and lower number = judge 3 · two separate judges, not averaged · each = mean of that memory's three answers

Damage zone — A false note led to silent failure (3 of 3). With healthy memory or with no memory: 0 of 3. With no memory, the agent said “I don't know” or stayed general, and broke nothing. **What was missing: a way to know where a fact came from and when it expires.**

How did we solve it?

The answer came much later, and it led us to build our own protection system. Take a sheet from the envelope, or ask me.

A small study: one model, three tasks, one answer each, no tools. The author was judge 1 and did not know which memory he was grading. It shows this can happen, not how often.

日本語 · 要約
誤った記憶（メモ）は、記憶がない場合より危険でした。古い情報により、AIは3回とも気づかずに誤答しました。記憶がない場合は、推測を事実として示しませんでした。

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“The agent remembered, but the world it lived in had changed.”